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Gene Expression and Biotechnology

Edexcel A A-Level Biology — Topic 8

8.1 Control of Gene Expression

Gene expression: The process by which information encoded in a gene is used to synthesise a functional gene product (usually a protein, but also functional RNA such as rRNA and tRNA). Although every nucleated cell in an organism contains the same genome, **differential gene expression** means that only a subset of genes is active in any given cell type. This is how a single fertilised egg gives rise to ~200 different cell types, each with a distinctive pattern of protein production. A liver cell and a neurone contain the same DNA but express different genes.

Levels of Gene Expression Control

Gene expression can be regulated at multiple levels, providing fine-tuned control over which proteins are produced, when, where, and in what quantities:

Level	Mechanism	Example	Additional Detail
Transcriptional (most important)	Transcription factors (activators/repressors) bind to promoter, enhancer, or silencer regions of DNA; determine whether RNA polymerase can access and transcribe the gene	Lac operon in bacteria; steroid hormone-receptor complexes acting as transcription factors in eukaryotes	The primary point of regulation in most organisms. Chromatin remodelling (see epigenetics) also affects accessibility
Post-transcriptional	Processing of pre-mRNA: alternative splicing (different	The Drosophila *Dscam* gene	Allows one gene to code for multiple

	combinations of exons retained in the mature mRNA); polyadenylation; mRNA stability and half-life	produces >38,000 different mRNA variants by alternative splicing. In humans, ~95% of multi-exon genes undergo alternative splicing	different proteins, greatly increasing proteome diversity beyond the ~20,000 genes
Translational	Regulation of translation initiation; mRNA degradation rate controlled by RNA-binding proteins; **microRNAs (miRNA)** bind to complementary mRNA sequences and either block translation or target mRNA for degradation (RNA interference)	Iron response elements in ferritin mRNA; lin-4 miRNA in *C. elegans* development	miRNAs are ~22 nucleotides long; estimated to regulate >60% of human genes. They are key players in development and cancer
Post-translational	Protein modification after synthesis: phosphorylation, glycosylation, acetylation, ubiquitination (tagging for degradation by proteasome); proteolytic cleavage to activate; protein folding by chaperones	Insulin: translated as preproinsulin → signal peptide cleaved → proinsulin → C-peptide cleaved → active insulin (2 chains linked by disulfide bonds)	Allows rapid, reversible control of protein activity without new transcription; ubiquitin-proteasome pathway essential for protein turnover

Transcription Factors

Transcription factors are proteins that bind to specific DNA sequences to regulate the rate of transcription. They are central to differential gene expression:

- **Activators:** Bind to enhancer sequences (which may be thousands of base pairs from the gene); help recruit RNA polymerase to the promoter and/or facilitate assembly of the transcription

initiation complex. They increase the rate of transcription

- **Repressors:** Bind to silencer sequences or directly to the promoter/operator; block RNA polymerase binding or prevent the initiation complex from forming. They decrease or prevent transcription
- **Basal transcription factors:** Required for all gene transcription; form the general transcription machinery with RNA polymerase at the TATA box (core promoter)

In eukaryotes, transcription factors work in **combinations** (combinatorial control), allowing a relatively small number of transcription factors to generate a vast number of distinct gene expression patterns. This is how ~2,000 transcription factor genes can control the expression of ~20,000 protein-coding genes in different cell types.

The Lac Operon (Prokaryotic Gene Regulation)

The lac operon in *E. coli* is a classic model of **inducible gene expression**, demonstrating how bacteria regulate gene expression in response to environmental conditions to conserve energy and resources:

Components:

- **Regulatory gene (*lacI*):** Constitutively expressed; produces the lac repressor protein
- **Promoter:** Where RNA polymerase binds to begin transcription
- **Operator:** A short DNA sequence between the promoter and structural genes; the repressor binding site
- **Structural genes:** *lacZ* (beta-galactosidase, cleaves lactose into glucose + galactose), *lacY* (permease, transports lactose into the cell), *lacA* (transacetylase)

Without lactose (genes OFF): The repressor protein (produced by *lacI*) binds tightly to the operator sequence. This physically blocks RNA polymerase from moving past the operator to transcribe the structural genes. No lactose-metabolising enzymes are produced. This saves energy — why make enzymes for a substrate that is not present?

With lactose present (genes ON): Lactose (or its isomer allolactose) acts as an **inducer**. It binds to the repressor protein, causing an allosteric conformational change that prevents the repressor from binding to the operator. With the operator free, RNA polymerase can proceed and transcribe *lacZ*, *lacY*, and *lacA* as a single polycistronic mRNA. The enzymes for lactose metabolism are produced.

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When lactose is used up: No more inducer is available → repressor returns to its active conformation → binds operator → transcription stops. This is an elegant, rapid, and reversible regulatory system.

> **Exam Tip:** The lac operon is an example of **negative inducible** control. "Negative" because the default regulatory mechanism is repression (a repressor blocks transcription). "Inducible" because the genes are normally OFF and are switched ON by the presence of the inducer (lactose). Contrast this with **negative repressible** systems like the trp operon, where genes are normally ON and switched OFF when the end product (tryptophan) accumulates. Examiners commonly ask you to explain the difference between inducible and repressible operons.

8.2 Epigenetics

> **Epigenetics:** The study of heritable changes in gene expression that do **NOT** involve changes to the underlying DNA base sequence. Epigenetic modifications alter the accessibility of DNA to the transcription machinery, effectively switching genes on or off without mutating the genetic code. These modifications can be influenced by environmental factors and, in some cases, passed to subsequent generations.

Mechanisms of Epigenetic Modification

Mechanism	How It Works	Effect on Gene Expression	Reversibility
DNA methylation	Methyl groups ($-CH_3$) added to cytosine bases by DNA methyltransferases, typically at **CpG islands** (regions rich in CG dinucleotides, often near promoters)	**Silences** gene expression: (1) methyl groups physically block transcription factor binding, (2) methylated DNA recruits methyl-binding proteins that in turn recruit histone	Reversible by demethylase enzymes; maintained through cell division by maintenance methyltransferases

		deacetylases, condensing the chromatin	
Histone acetylation	Acetyl groups added to lysine residues on histone tails by **histone acetyltransferases (HATs)**	**Activates** gene expression: acetylation neutralises the positive charge on histones, weakening their electrostatic attraction to negatively charged DNA. Chromatin loosens into open **euchromatin**, making DNA accessible to RNA polymerase and transcription factors	Reversed by histone deacetylases (HDACs), which remove acetyl groups → chromatin condenses into **heterochromatin** → gene silenced
Histone methylation	Methyl groups added to histone tails by histone methyltransferases	Can either activate or silence genes depending on which amino acid is methylated and how many methyl groups are added (mono-, di-, or tri-methylation)	Reversed by histone demethylases

Significance and Applications of Epigenetics

- **Cell differentiation:** Explains how all cells share the same genome yet become specialised. During development, epigenetic marks are progressively added to silence genes not needed in a particular cell lineage (e.g. haemoglobin genes are unmethylated in red blood cell precursors but heavily methylated in neurones)
- **X-chromosome inactivation:** In female mammals, one X chromosome in each cell is almost entirely silenced by DNA methylation and histone modification, forming a Barr body. This ensures dosage compensation between XX females and XY males
- **Genomic imprinting:** Certain genes are expressed from only one parental allele (either maternal or paternal), controlled by parent-specific methylation patterns established during gamete formation

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- **Environmental influence:** Diet (e.g. folate provides methyl groups), stress hormones, toxins (e.g. bisphenol A), exercise, and ageing all influence epigenetic marks. This provides a mechanism by which the environment can alter gene expression without mutating DNA
- **Cancer:** Abnormal epigenetic patterns are a hallmark of cancer. Hypermethylation of tumour suppressor gene promoters silences them; global hypomethylation can activate oncogenes. Epigenetic drugs (HDAC inhibitors, demethylating agents) are used in cancer therapy
- **Transgenerational inheritance:** Some epigenetic marks survive reprogramming during gamete formation and can be passed to offspring. Studies on the Dutch Hunger Winter (1944–45) showed that children and grandchildren of famine-exposed mothers had altered metabolic profiles, suggesting epigenetic inheritance
- **Twin studies:** Monozygotic twins have identical genomes but accumulate epigenetic differences over their lifetimes (epigenetic drift), explaining why they may differ in disease susceptibility, personality, and appearance as they age

Exam Tip: Key distinction that examiners test: **mutations** change the DNA base sequence (permanent, irreversible under normal conditions); **epigenetic changes** modify gene expression without altering the sequence (potentially reversible with specific enzymes or drugs). Both can be heritable and both can contribute to disease. Do not describe epigenetic changes as mutations — this will lose marks.

8.3 Gene Technology Tools

Restriction Enzymes (Restriction Endonucleases)

Restriction enzyme: An enzyme (originally isolated from bacteria, where they provide defence against bacteriophage DNA) that cuts double-stranded DNA at or near specific **palindromic recognition sequences** (4–8 base pairs long, reading the same on both strands in the 5' to 3' direction). There are thousands of different restriction enzymes, each recognising a different sequence.

Types of cut:

- **Sticky ends:** The enzyme cuts the two strands at different positions, leaving short single-stranded overhangs (typically 2–4 bases). Example: *Eco*RI cuts GAATTC to produce 4-base 5' overhangs. Sticky ends are preferred in genetic engineering because complementary sticky ends from different DNA sources can form hydrogen bonds (base pair) and be easily joined by DNA ligase
- **Blunt ends:** The enzyme cuts both strands at the same position. Harder to ligate but more versatile (any blunt end can join to any other). Example: *Sma*I cuts CCCGGG to produce blunt ends

Key principle: If the same restriction enzyme is used to cut both the gene of interest and the vector (plasmid), they will have **complementary sticky ends** that can anneal, allowing recombinant DNA to be created.

DNA Ligase

An enzyme that catalyses the formation of **phosphodiester bonds** between the sugar-phosphate backbones of DNA fragments, permanently sealing the nicks. In genetic engineering, DNA ligase is used to join the gene of interest into the vector after restriction enzyme cutting. T4 DNA ligase (from bacteriophage T4) is most commonly used in the laboratory.

Reverse Transcriptase

An enzyme (originally from retroviruses such as HIV) that synthesises **complementary DNA (cDNA)** from an mRNA template. This is crucial when you want to clone a eukaryotic gene but need a version without introns (since bacteria cannot splice introns from pre-mRNA). The process: mRNA (already spliced, containing only exons) is isolated from cells expressing the gene of interest → reverse transcriptase produces a single-stranded cDNA copy → DNA polymerase produces the complementary second strand → double-stranded cDNA can be inserted into a bacterial vector.

> **Exam Tip:** A common exam question asks why cDNA is used rather than genomic DNA when inserting a eukaryotic gene into bacteria. Answer: (1) bacteria lack the splicing machinery (spliceosomes) to remove introns from pre-mRNA, so the gene must be intron-free; (2) cDNA made from mRNA contains only exons; (3) cDNA represents only expressed genes, making it easier to find the gene of interest from cells that actively produce the target protein.

Polymerase Chain Reaction (PCR)

PCR: An *in vitro* technique for rapidly amplifying (copying) a specific DNA sequence, producing millions of identical copies from a tiny starting sample in just a few hours. Invented by Kary Mullis in 1983 (Nobel Prize 1993). Requirements: (1) template DNA (even nanogram quantities), (2) two specific **primers** (short synthetic oligonucleotides, ~18–25 bases, complementary to sequences flanking the target region), (3) **Taq DNA polymerase** (thermostable, from *Thermus aquaticus*, a bacterium living in hot springs at ~70–80 degrees C), (4) free deoxyribonucleotide triphosphates (dNTPs: dATP, dTTP, dCTP, dGTP), (5) buffer solution containing Mg^{2+} (cofactor for Taq polymerase).

The PCR Cycle (3 Steps, Repeated 25–35 Times)

1. **Denaturation (94–98 degrees C, ~30 s):** High temperature breaks the hydrogen bonds between complementary base pairs; double-stranded DNA separates completely into two single strands. This temperature would denature most enzymes, but Taq polymerase is thermostable
2. **Annealing (50–65 degrees C, ~30 s):** Temperature is lowered to allow the primers to bind (anneal) to their complementary sequences on the template strands by hydrogen bonding. The exact annealing temperature depends on the primer sequence (longer primers and those with higher GC content require higher temperatures). Primers bind before the template strands can re-anneal because primers are present in vast excess
3. **Extension/Elongation (72 degrees C, 1–2 min):** The optimum temperature for Taq polymerase. The enzyme extends each primer by adding complementary dNTPs in the **5' to 3' direction**, synthesising new DNA strands complementary to the template. Extension time depends on the length of the target sequence (~1 min per kb)

After n cycles, the number of copies = $\mathbf{2^n}$ (exponential amplification). After 30 cycles, a single molecule is amplified to over 1 billion copies.

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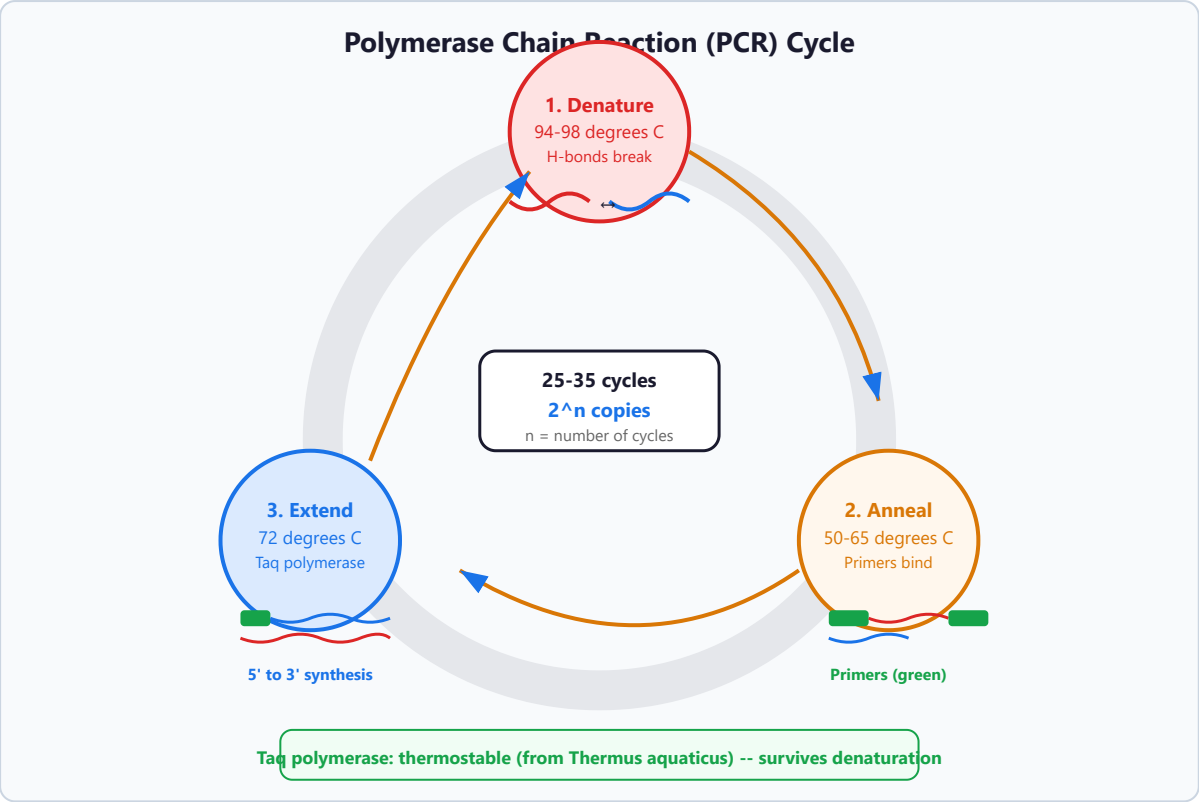


Number of DNA copies = 2^n (where n = number of cycles). After 30 cycles: 2^{30} = 1,073,741,824 copies.

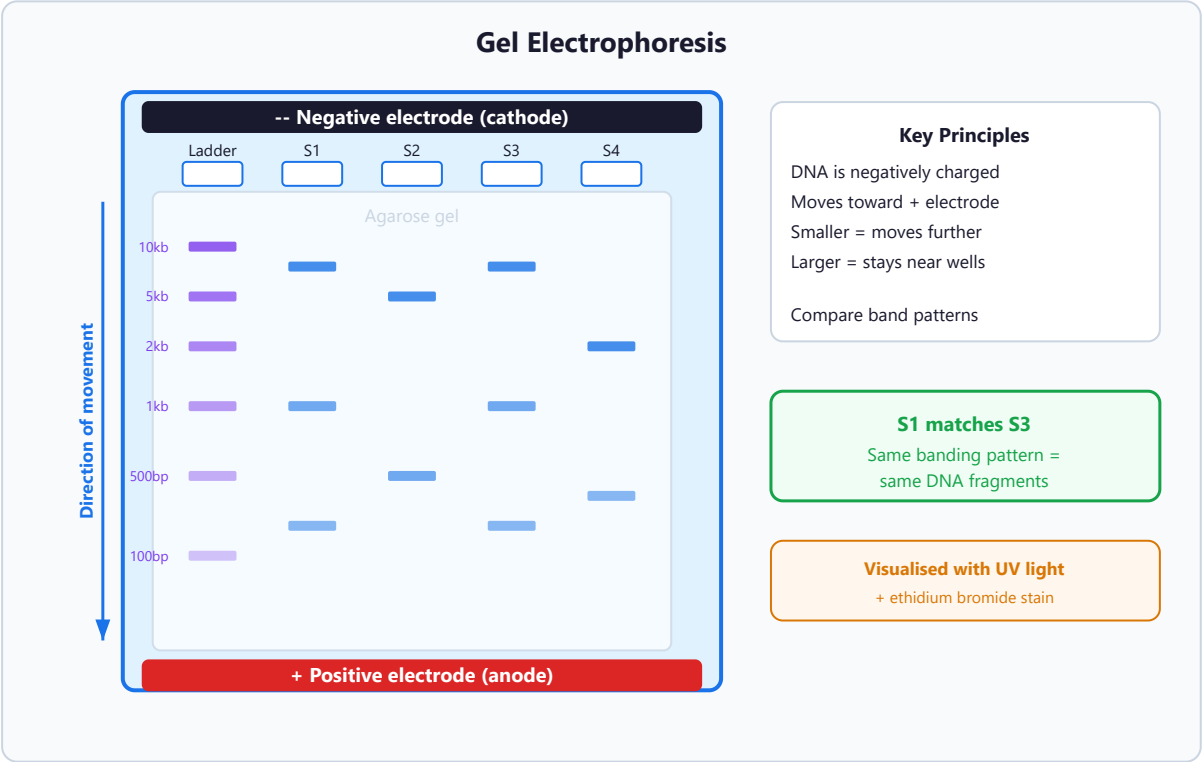
Applications of PCR

- **Forensic DNA profiling** — amplifying tiny crime scene samples (a single hair root contains enough DNA)
- **Medical diagnosis** — detecting viral DNA/RNA (e.g. COVID-19 RT-PCR tests; detecting HIV proviral DNA)
- **Genetic testing** — amplifying specific alleles to test for genetic diseases (e.g. cystic fibrosis, sickle cell)
- **Palaeontology** — amplifying ancient DNA from fossils or preserved specimens
- **Paternity testing and ancestry analysis**
- **Cloning genes** into vectors for genetic engineering

Gel Electrophoresis

A technique that separates DNA (or protein) fragments by size and charge using an electric field through a gel matrix:

1. DNA samples mixed with loading dye are pipetted into wells at one end of an **agarose gel** (for DNA) or polyacrylamide gel (for proteins) submerged in buffer solution
2. An electric current is applied across the gel. DNA is **negatively charged** (phosphate groups) and migrates towards the **positive electrode (anode)**
3. The gel acts as a molecular sieve: **smaller fragments migrate faster and further** because they encounter less frictional resistance passing through the gel pores; larger fragments are retarded
4. After separation, fragments are visualised using UV light with ethidium bromide or SYBR Green stain (intercalates into DNA and fluoresces), or by fluorescent labelling
5. A **DNA ladder** (size markers of known fragment lengths) is run alongside samples to estimate fragment sizes by comparison



8.4 Genetic Engineering

Genetic engineering (recombinant DNA technology): The deliberate modification of an organism's genome by isolating a gene from one organism (or synthesising it artificially) and

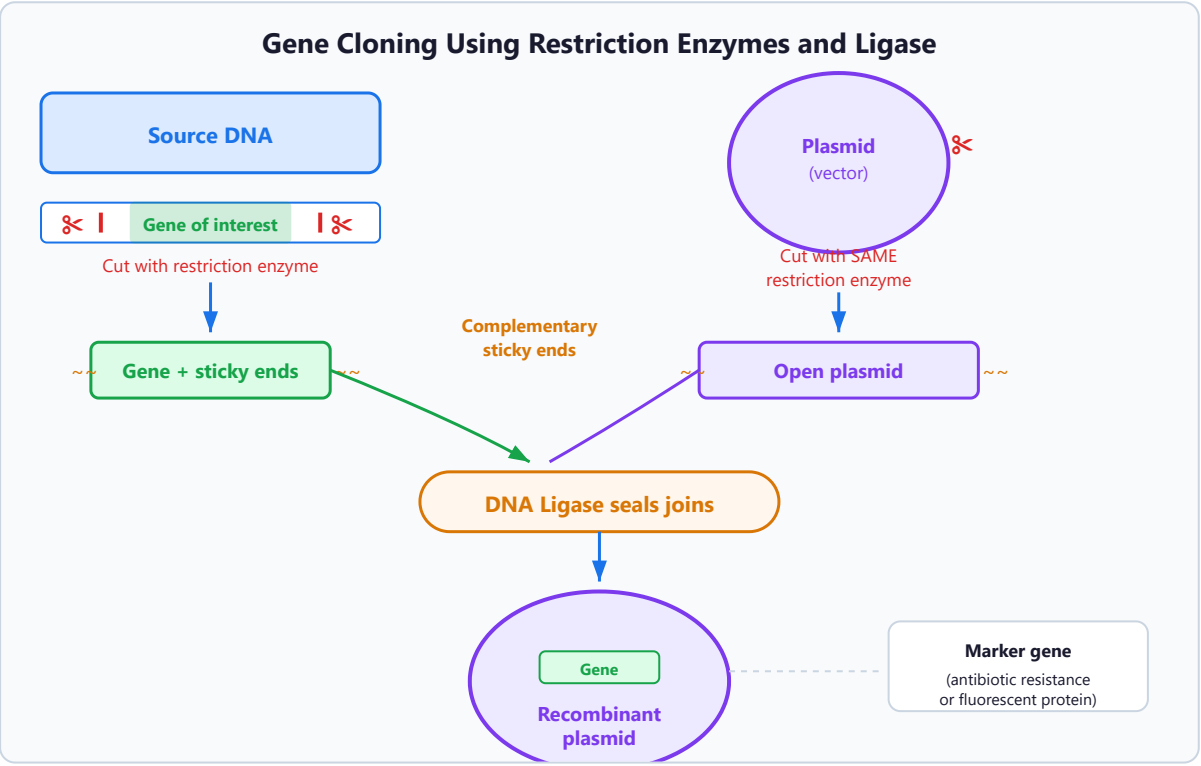
introducing it into another organism (often of a different species), producing a **transgenic** or **genetically modified organism (GMO)**. The inserted gene is called a **transgene** and the organism expressing it is the **transgenic organism**.

Steps in Genetic Engineering (Overview)

1. **Identify and isolate the gene of interest.** Three main approaches:
   - **Restriction enzymes:** Cut the gene from genomic DNA using known flanking restriction sites
   - **Reverse transcriptase:** Isolate mRNA from cells expressing the protein → produce cDNA (complementary DNA, intron-free). This is essential for eukaryotic genes being expressed in prokaryotes
   - **Chemical synthesis:** If the amino acid sequence of the protein is known, the gene can be synthesised artificially using a DNA synthesiser. Allows codon optimisation for the host organism
2. **Insert the gene into a vector.** Common vectors include:
   - **Bacterial plasmids:** Small circular DNA molecules; contain an origin of replication, antibiotic resistance gene(s) as selectable markers, and multiple cloning sites (MCS). Cut with the same restriction enzyme as the gene → complementary sticky ends → DNA ligase seals the joins → recombinant plasmid
   - **Bacteriophages:** Viral vectors; can carry larger inserts
   - **Artificial chromosomes (BACs, YACs):** For very large DNA inserts
3. **Transform the host cell** (introduce recombinant vector):
   - **Heat shock:** Brief exposure to 42 degrees C makes bacterial membranes temporarily permeable to DNA
   - **Electroporation:** Brief electric pulse creates transient pores in the membrane
   - **Microinjection:** Direct injection of DNA into cell nucleus using fine glass pipette
   - **Viral infection:** Modified virus delivers gene to animal cells
   - **Gene gun (biolistics):** DNA-coated gold particles fired into plant cells at high velocity
   - **Agrobacterium tumefaciens:** Natural plant pathogen modified to carry the desired gene; inserts DNA into plant chromosome via Ti plasmid
4. **Identify successfully transformed cells** using **selectable marker genes**:

- **Antibiotic resistance:** Grow cells on agar containing antibiotic; only cells with the plasmid (carrying resistance gene) survive
- **Fluorescent markers (e.g. GFP):** Transformed cells glow under UV light
- **Insertional inactivation:** Gene of interest inserted into a second marker gene, disrupting it. Cells with recombinant plasmid lose that second marker (e.g. blue/white screening with lacZ)

5. **Clone and express the gene.** Grow transformed cells in industrial fermenters; the transgene is transcribed and translated to produce the desired protein, which is then purified



Applications of Genetic Engineering

Application	Example	Benefit	Detail

Medicine	Human insulin produced by GM *E. coli* or yeast (*S. cerevisiae*)	Unlimited supply; no allergic reactions (identical to human protein); pure and consistent; cheaper than animal-derived insulin	Before 1982, insulin was extracted from pig and cow pancreases, which sometimes caused immune reactions. The human insulin gene was inserted into bacterial plasmids, revolutionising diabetes treatment
Agriculture	Bt crops (gene from *Bacillus thuringiensis* produces Cry protein insecticidal toxin)	Reduced pesticide use; increased yield; less environmental damage from spraying	Bt toxin is specific to certain insect larvae (lepidopterans); it disrupts their gut membranes. Bt cotton and Bt maize are widely grown
Agriculture	Golden Rice (beta-carotene biosynthesis genes from daffodil and *Erwinia* bacterium)	Addresses vitamin A deficiency (affects 250 million children globally); beta-carotene converted to vitamin A in the body	Controversial: critics argue it distracts from broader poverty and dietary diversity solutions; supporters say it is a simple, effective intervention
Agriculture	Herbicide-resistant crops (Roundup Ready soya)	Allows broad-spectrum herbicide use without crop damage; simplifies weed management	Concerns about herbicide-resistant weeds evolving through gene flow
Research	GFP (green fluorescent protein) as reporter gene	Tracks gene expression in real time; visualises protein localisation in living cells	GFP from jellyfish *Aequorea victoria*; Nobel Prize to Shimomura, Chalfie, Tsien (2008)
Medicine	Genetically engineered vaccines (e.g. hepatitis B surface	Safer than using whole pathogens; consistent production	The antigen gene is expressed in yeast; purified antigen used in vaccine

	antigen produced in yeast)		

Cloning Organisms

Reproductive cloning (somatic cell nuclear transfer, SCNT):

1. An egg cell is **enucleated** (nucleus removed by micropipette)
2. A **diploid nucleus** from an adult somatic cell (e.g. udder cell) is inserted into the enucleated egg
3. The reconstructed cell is stimulated (electric shock) to trigger mitotic division
4. The embryo develops to blastocyst stage and is implanted into a surrogate mother
5. The offspring is a **genetic clone** of the nuclear donor (identical nuclear DNA, but mitochondrial DNA comes from the egg donor)

Example: **Dolly the sheep** (1996) — first mammal cloned from an adult somatic cell. She developed normally but showed premature ageing (arthritis, shortened telomeres) and was euthanised at age 6. Success rates for SCNT remain very low (~1–3%).

Therapeutic cloning: The same nuclear transfer technique, but the embryo is not implanted. Instead, embryonic stem cells are harvested from the blastocyst (inner cell mass). These pluripotent stem cells can be directed to differentiate into specific tissue types for transplantation, with the advantage of being genetically matched to the patient (no immune rejection). This remains ethically controversial due to the destruction of embryos.

Ethical, Social and Ecological Issues of Genetic Engineering

Concern	Detail
Gene flow to wild populations	Transgenes could spread via pollen to wild relatives, creating herbicide-resistant "superweeds" or disrupting wild ecosystems
Loss of biodiversity	Monocultures of GM crops reduce genetic diversity; traditional varieties may be abandoned
Unknown long-term effects	Insufficient data on long-term health effects of consuming GM food (though no evidence of harm after 25+ years of use)

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Corporate control	GM seeds are patented; farmers cannot save and replant seeds; dependency on multinational corporations
Ethical / religious objections	Concerns about "playing God" or crossing species boundaries; some view modification of life as inherently wrong
Unequal access	Benefits may not reach those who need them most (LIDCs); technology concentrated in wealthy nations
Precautionary principle	Some argue we should not proceed until we are certain of safety, even if current evidence suggests no harm

Exam Tip: When discussing GM organisms in extended-answer questions, always present a **balanced argument** with both benefits AND risks. Use specific named examples (Golden Rice, Bt cotton, insulin). Conclude by evaluating the weight of evidence on each side. Examiners reward nuanced answers that acknowledge complexity rather than one-sided arguments.

8.5 Biotechnology and Fermentation

Biotechnology: The use of living organisms, or their biological systems and products, in industrial processes to manufacture useful substances or provide services. It spans ancient practices (bread, beer, cheese) to modern applications (gene therapy, bioremediation, industrial enzyme production). The organisms used are typically microorganisms (bacteria, fungi, algae) because they reproduce rapidly, are easy to culture, and can be genetically modified.

Industrial Fermenters (Bioreactors)

Large stainless steel vessels (up to 500,000 litres) designed to maintain optimal conditions for microbial growth and product formation:

Condition	How Controlled	Why Important

Temperature	Water-cooled jacket or internal cooling coils; temperature probes linked to computer control	Metabolic reactions are exothermic; excess heat would denature enzymes. Optimal temperature maintained (e.g. 37 degrees C for *E. coli*, 25–30 degrees C for *Penicillium*)
pH	pH probe; automatic addition of acid or alkali (buffers)	Enzyme activity is pH-dependent; microbial metabolites (organic acids, CO_2) can alter pH. Maintained at species-specific optimum
Oxygen (for aerobic organisms)	Sterile air supply (filtered through bacterial filters); mechanical agitator/impeller	Aerobic respiration required for maximum growth and product yield; agitator ensures even distribution of O_2, nutrients, and heat
Nutrients	Sterile nutrient medium containing carbon source (glucose, sucrose), nitrogen source (ammonium salts, amino acids), minerals, vitamins	Essential for growth and product synthesis; composition optimised for specific organism and product
Aseptic conditions	Vessel sterilised with steam under pressure (autoclaved) before use; medium sterilised; air filtered; system sealed; sampling ports sterilised	Prevents contamination by unwanted microorganisms that would compete for nutrients, produce toxins, or degrade the desired product

Batch vs Continuous Culture

Feature	Batch Culture	Continuous Culture
Process	Fixed volume of sterile medium inoculated with organism; nothing added or removed until end. Products harvested when culture is complete	Fresh medium continuously added; culture broth and products continuously removed at the same rate. System reaches a **steady state**

Growth phases	All phases observed: lag (adaptation), log/exponential (rapid growth), stationary (nutrients depleted, waste accumulates), death (decline)	Maintained indefinitely in log (exponential) phase by constant nutrient supply and waste removal
Productivity	Higher product concentration at end; downtime between batches for cleaning and re-sterilisation	Lower concentration but constant output; higher overall productivity per unit time
Contamination risk	Lower (closed system; short runs; easy to sterilise between batches)	Higher (open system with continuous input/output; long runs; any contamination affects entire production)
Flexibility	Different products can be made in successive batches	Dedicated to one product for extended periods
Examples	Penicillin production, beer/wine, insulin (GM bacteria), yoghurt	Mycoprotein (Quorn from *Fusarium venenatum*), single-cell protein, waste-water treatment

8.6 Gene Therapy

Gene therapy: The introduction of a normal (functional) copy of a gene into the cells of a patient to treat or prevent disease caused by a defective or absent gene. The aim is to correct the genetic defect at the molecular level, either by replacing the faulty gene, supplementing it with a working copy, or silencing a harmful gene. Gene therapy primarily targets **single-gene (monogenic) disorders**.

Types of Gene Therapy

Type	Target Cells	Heritable?	Current Status	Ethical Position

Somatic cell gene therapy	Body cells of the affected individual (e.g. lung epithelial cells, bone marrow cells, liver cells)	No — genetic change not passed to offspring; dies with the patient	In clinical use for some conditions (e.g. SCID, some cancers, spinal muscular atrophy, sickle cell disease)	Generally accepted ethically (similar to any other medical treatment)
Germ line gene therapy	Gametes (sperm/egg) or early embryo	Yes — genetic change passed to all future generations	Currently **banned** in humans in most countries	Highly controversial: benefits of eliminating genetic disease vs risks of unintended heritable changes, consent of future generations, potential for non-medical "enhancement" (designer babies)

Delivery Vectors for Gene Therapy

Vector	Mechanism	Advantages	Disadvantages
Viral vectors (adenovirus, retrovirus, lentivirus, AAV)	Therapeutic gene inserted into modified viral genome; virus infects target cells and delivers gene. Virus modified to be replication-deficient (cannot cause disease)	High efficiency of gene delivery; natural cell-entry mechanism; some integrate into host genome for long-term expression (retrovirus, lentivirus)	Immune response against viral proteins; risk of **insertional mutagenesis** (gene integrates near an oncogene, potentially causing cancer); limited insert size; difficulty in large-scale production
Liposomes	Lipid vesicles encapsulating the therapeutic DNA; fuse with target cell	Less immunogenic than viruses; no risk of insertional mutagenesis; can	Lower efficiency of gene delivery and expression; gene expression often

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	membrane, releasing DNA into cytoplasm	carry larger DNA inserts; relatively safe	transient; targeted delivery difficult
Direct injection / electroporation	Naked DNA injected directly into tissue or cells made permeable by electric pulse	Simple; no vector-related immune response	Very low efficiency; DNA rapidly degraded by nucleases; not practical for most tissues

Challenges and Limitations

- **Targeting correct cells:** Difficult to deliver the gene specifically to the affected tissue (e.g. brain cells are hard to reach)
- **Immune response:** The body may mount an immune response against the vector (especially viral), reducing effectiveness on repeated treatment and potentially causing dangerous inflammation
- **Insertional mutagenesis:** If the transgene integrates randomly into the host chromosome, it may disrupt a tumour suppressor gene or activate an oncogene, potentially causing cancer
- **Temporary effect:** Somatic cells divide; the transgene may be diluted out over time, requiring repeated treatments. Non-integrating vectors (adenovirus) require ongoing administration
- **Limited to monogenic disorders:** Most diseases involve multiple genes (polygenic) and/or environmental factors, making gene therapy impractical
- **Cost:** Gene therapies are extremely expensive (e.g. Zolgensma costs over $2 million per treatment), raising issues of access and equity

8.7 Genomics and Bioinformatics

Genomics: The study of the entire genome of an organism, including the mapping, sequencing, and functional analysis of all its DNA. The **Human Genome Project (HGP)**, completed in 2003 after 13 years and $3 billion, sequenced all ~3.2 billion base pairs of human DNA, identifying approximately 20,000–25,000 protein-coding genes. This was a landmark achievement that has transformed biology and medicine.

Key Findings of the Human Genome Project

- Humans have only ~20,000–25,000 protein-coding genes (far fewer than the ~100,000 predicted), accounting for only ~1.5% of the genome
- ~98.5% of the genome is non-coding ("junk" DNA), though much of this has regulatory functions (enhancers, promoters, non-coding RNAs, transposable elements)
- Humans share ~99.9% of their DNA sequence with each other; the ~0.1% variation accounts for all individual differences
- We share ~98.8% of DNA with chimpanzees, ~85% with mice, and ~60% with bananas, revealing deep evolutionary conservation
- Many genes are highly conserved across species, confirming common ancestry

DNA Sequencing Methods

Sanger sequencing (chain termination method, 1977): Uses dideoxynucleotides (ddNTPs) that lack the 3'-OH group needed for the next phosphodiester bond, terminating DNA synthesis at random positions. Produces fragments of every possible length, separated by capillary electrophoresis. Each ddNTP is fluorescently labelled with a different colour, allowing automated reading of the sequence. Still used as the "gold standard" for validating short sequences.

Next-generation sequencing (NGS): Massively parallel sequencing technologies that can sequence millions of DNA fragments simultaneously. Orders of magnitude faster and cheaper than Sanger (a human genome now costs ~$200 and takes hours, compared to years and billions of dollars). Examples: Illumina (sequencing by synthesis), Oxford Nanopore (real-time, portable). NGS has made whole-genome sequencing routine in research and increasingly in clinical medicine.

Bioinformatics

The use of computer science, mathematics, and statistics to store, manage, analyse, and interpret the vast quantities of biological data generated by genomics, proteomics, and other "-omics" disciplines. Essential applications include:

- **Sequence alignment and comparison:** BLAST (Basic Local Alignment Search Tool) searches databases to find similar sequences across species, identifying homologous genes and inferring function

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- **Gene annotation:** Identifying the locations, boundaries, and functions of genes within raw sequence data using computational prediction and comparison with known genes
- **Phylogenetics:** Constructing evolutionary trees from DNA/protein sequence comparisons; determining how closely related species or populations are
- **Protein structure prediction:** Using amino acid sequences to predict 3D protein structures (AlphaFold by DeepMind achieved breakthrough accuracy in 2020)
- **Personalised medicine / pharmacogenomics:** Analysing an individual's genome to predict drug responses, adverse effects, and disease susceptibility, enabling tailored treatment plans

Applications of Genomics

Application	Description	Example
Personalised/precision medicine	Drug selection and dosage based on individual genetic profile	HER2 testing determines eligibility for Herceptin in breast cancer; warfarin dosing adjusted based on CYP2C9 genotype
Genome-wide association studies (GWAS)	Scanning entire genomes of large populations to identify genetic variants associated with diseases	Identified >100 loci associated with type 2 diabetes risk
Comparative genomics	Comparing genomes across species to understand evolution, gene function, and conservation	Mouse knockout studies informed by human-mouse gene homology
Metagenomics	Sequencing DNA directly from environmental samples without culturing organisms	Characterising the human gut microbiome; analysing soil biodiversity
Synthetic biology	Designing and constructing new biological systems, pathways, or organisms from scratch	Synthetic yeast genome project (Sc2.0); engineered bacteria that produce artemisinin (antimalarial drug)

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CRISPR-Cas9 Gene Editing

CRISPR-Cas9 (Clustered Regularly Interspaced Short Palindromic Repeats) is a revolutionary gene editing technology adapted from a natural bacterial immune defence system against viruses. It allows precise, targeted modification of DNA sequences in virtually any living organism:

1. A **guide RNA (gRNA)** of ~20 nucleotides is designed to be complementary to the specific target DNA sequence
2. The gRNA forms a complex with the **Cas9 endonuclease** protein and directs it to the exact target location in the genome by complementary base pairing
3. Cas9 creates a precise **double-strand break (DSB)** in the DNA at the target site
4. The cell's own DNA repair mechanisms fix the break in one of two ways:
   - **Non-homologous end joining (NHEJ):** Error-prone repair that often introduces insertions or deletions (indels), disrupting the gene → gene knockout (loss of function)
   - **Homology-directed repair (HDR):** If a DNA template with the desired sequence is provided, the cell uses it as a template to repair the break, allowing precise gene correction or insertion of new sequences

Advantages of CRISPR-Cas9 Over Traditional Genetic Engineering

- **Precision:** Can target any specific sequence in the genome (just design the appropriate gRNA)
- **Speed:** Experiments that took months with older techniques can be done in weeks
- **Cost:** Far cheaper than previous methods (zinc finger nucleases, TALENs)
- **Versatility:** Works in virtually all cell types and organisms tested
- **In situ editing:** Modifies genes in their natural chromosomal location (no need to remove and reinsert)
- **Multiplexing:** Can target multiple genes simultaneously using different gRNAs

> **Exam Tip:** For any biotechnology or genomics question, be prepared to discuss **ethical, social and ecological implications**. Key points to include: informed consent, equitable access, unintended consequences (off-target edits), environmental release of edited organisms, designer babies and genetic enhancement vs therapy, the distinction between somatic and germ line editing, patent and ownership issues, and the precautionary principle vs potential benefits. A well-structured evaluation considering multiple perspectives will score highly.

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